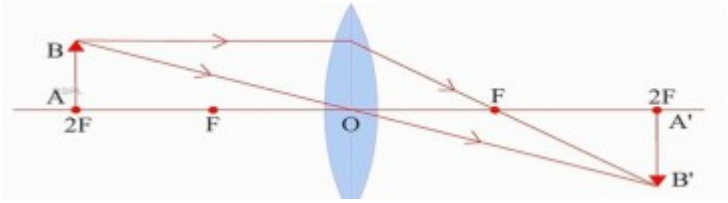
Question No.	SSLC MODEL EXAMINATION 2026 PHYSICS – ANSWER KEY -EM	Score
1	**c.** Both the statement and the reason are correct, and the reason explains the statement.	
2	**a.** (ii) only.	
3	**d.** (1) ii, (2) iv, (3) i, (4) iii.	
4	**c.** (i), (ii), (iv).	
5 a.	Distance = Speed X Time= 350 X 0.02 = 7m	
b.	Echo cannot be heard here because the time taken for the sound to reflect (0.04s) is less than the 0.1s required for the human ear to distinguish distinct sounds. Minimum distance required for an echo is 17.5m	
6 a.	Concave lens.	
b.	25cm (Focal length =1/Power) =1/4=0.25m	
7	Joule's Law: Energy loss as heat occurs during transmission. Solution: Use low-resistivity wires or increase voltage to decrease current.	
8 a.	1.	
b.	Load arm = Pulley radius (5 cm); Effort arm = Pulley diameter (10 cm).	
9 A. a.	From B to A.	
b.	**Right-Hand Thumb Rule:** If a conductor is held in the right hand with the thumb in the direction of the current, the other fingers show the magnetic field direction.	
9 B. a.	From A to B.	
b.	A magnetic field is formed around a current-carrying conductor.	
10	$H = V^2 / (R\ t)$ $R = V^2 / (H\ t)$ $= 200 \times 200 / (12000 \times 1 \times 60)$ $= 200\ \Omega$	

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11 a.	Size of eyeball is small or lens power is low.	
b.	Hypermetropia; corrected using a convex lens.	
12 a.	A . Since it is an erect image, it will be virtual.	
b.		
13 A. a.	Red, Green, and Blue are primary colours because they can create all other colors.	
b.	Red, Green, Blue (any two).	
c.	Blue is a complementary color (Red + Green = Yellow; Yellow + Blue = White).	
13 B. a.	Yellow filter allows yellow, red, and green light. Leaves appear green and flowers yellow.	
b.	In green light, the flowerpot appears dark; leaves and flowers appear green.	
14	Reduce domestic energy use, don't waste food, use public transport, use recyclable products.	
15 A. a.	W = F .s = 400 N x 3 m = 1200 J	
b.	b.$E=(L\times h)/l=(400\times 3)/6=200N$.	
B. a.	Pitch =30cm/10=3cm	
b.	Mechanical Advantage= $\frac{\text{length of one Thread(l)}}{\text{Pitch(h)}}$	
c.	=15cm/3cm=5 Effort =10000/5=2000N	
16 a.	(a) .	
b.	If the armature doesn't rotate, the split-ring commutator doesn't rotate, producing AC in the external circuit.	

Question No.	SSLC MODEL EXAMINATION 2026 PHYSICS – ANSWER KEY -EM	Score
17 a.	From A to B.	
b.	Change current direction or place conductor above North Pole.	
18 A. a.	Longitudinal wave; requires a medium.	
b.	Distance between two consecutive compressions or rarefactions.	
c.	Velocity = Distance /Time=21m/2s = 10.5m/s	
B. a.	B.	
b.	Frequency of A = 3/12=0.25 Hz, B = 4/12 =0.33 Hz	
c.	Velocity A= 12m/12s = 1m/s ; Velocity B = 24m/12s=2m/s	

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